

Test
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sec-B

Rollno-21

① Node.js - event loop
console.log('A');
setTimeout(() => console.log('B'), 0);
promise.resolve().then(() =>
 console.log('C');
 console.log('D');

② Node.js - calc.js

```
function add(a,b){  
  return a+b;  
}
```

```
function subtract(a,b){  
  return a-b;  
}
```

```
module.exports = {  
  add, subtract  
};
```

```
const { add, subtract } =  
  require('./calc');  
console.log(add(4,5));
```

③ Middle ware

```
function requireAuth(req, res, next){  
  if (req.headers['x-api-key'] !== 'secret123')  
    return  
  res.status(401).json({ error: 'Unauthorized' });  
}
```

next():

```
app.get('/api/books',  
(req, res) => res.json(books));  
app.post('/api/books', requireAuth,  
(req, res) => {  
  res.status(201).json(req.body);  
});
```

④ REST App

```
app.get('/api/books', (req, res) => {  
  res.status(200).json(books);  
});
```

```
app.post('/api/books', (req, res) => {  
  const book = { id: books.length + 1, ...req.body };  
  books.push(book);  
  res.status(201).json(book);  
});
```

```
app.delete('/api/books/:id',  
(req, res) => {  
  const i = books.findIndex(b =>  
    b.id === req.params.id);
```

```
  if (i === -1) return;
```

```
  res.status(404).json({ error: 'Book  
  not found' });
```

```
  books.splice(i, 1);
```

```
  res.status(200).json({ message: 'Deleted' });  
});
```

⑤ Error handling

```
app.get('/api/books/:id',  
  (req, res) => {  
    const book = books.find(b =>  
      b.id === req.params.id);  
    if (!book)  
      return  
      res.status(404).json({ error: 'book  
      not found' });  
    res.json(book.title);  
  });
```